

MEMO

Sample test

QUESTION 1 [4 marks]

Solve for x .

i) $2 \ln x = \ln(2x - 1)$.

$$x > 0 \text{ and: } 2x - 1 > 0$$

$$2x > 1$$

$$x > \frac{1}{2}$$

$$\Rightarrow \boxed{x > \frac{1}{2}}$$

$$\ln x^2 = \ln(2x - 1)$$

$$\Rightarrow e^{\ln x^2} = e^{\ln(2x - 1)}$$

$$x^2 = 2x - 1$$

$$\Rightarrow x^2 - 2x + 1 = 0$$

$$(x - 1)^2 = 0$$

$$\Rightarrow \underline{\underline{x = 1}}$$

you can do this because
the exponential function
is one-to-one.

ii) $1 < e^{2x-2} < 2$.

$$\ln 1 < \ln e^{2x-2} < \ln 2$$

$$\Rightarrow 0 < 2x - 2 < \ln 2$$

$$2 < 2x < \ln 2 + 2$$

$$1 < x < \frac{1}{2} \ln 2 + 1$$

$$x \in (1, \frac{1}{2} \ln 2 + 1)$$

$$\text{or } x \in (1, \ln \sqrt{2} + 1)$$

(you can do this
because $\ln$ is
an increasing
function)

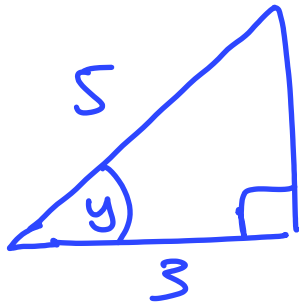
QUESTION 2 [3 marks]

Find the exact value of

$$\sin\left(2 \arccos \frac{3}{5}\right).$$

$$\text{let } y = \arccos \frac{3}{5}$$

$$\cos y = \frac{3}{5} \quad \text{and} \quad y \in [0, \pi]$$



$$\sqrt{25-9} = \sqrt{16} = 4$$

$$\begin{aligned} \sin\left(2 \arccos \frac{3}{5}\right) &= \sin 2y \\ &= 2 \sin y \cos y \\ &= 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) \\ &= \frac{24}{25}. \end{aligned}$$

QUESTION 3 [1 mark]

The displacement (in meters) of a particle at time t (in seconds) is given by $y(t) = t^2 + 4t$. Find the average velocity of the particle on the interval $[1, 2]$.

$$\begin{aligned} \text{Average velocity} &= \frac{y(2) - y(1)}{2 - 1} \\ &= \frac{(4 + 8) - (1 + 4)}{1} \\ &= \frac{12 - 5}{1} \\ &= 7 \text{ m/s} \end{aligned}$$